(¥)

Comparison

	Write (C,U,D)	Read	Cons.
TTL (Write Around) Eventually consistent	DB	① Cache ② DB	Stale data for sometime.
Write Through (Always) Consistent	Cache + DB	11	Write becomes slow
Write Back Inconsistency	Cache	11	Data (write) may get lost forever.

Eventual Consistency ≠ Consistency.

↓ w/ time, it will be consistent

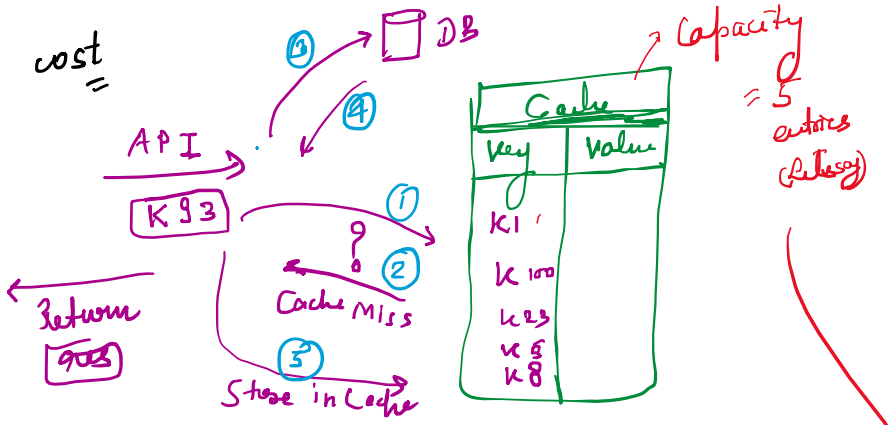
↓ consistent from the very moment

(*) How to handle Cache getting Full?

Cache - mem <<< 2° storage

Lesser cost ③ → DB Capacity

Reason cost =



Cache Miss → Cache is not having the data we are looking for

Cache Hit → Cache IS Having ... ...

When 6th entry needs to be inserted into the cache

Option #1 ✗ Ignore

Option #2 ✓ Remove an existing entry from Cache

What entry to remove

Cache Eviction Strategies

↳ Algo to decide what data to remove from cache when cache is full.

Eg → 1.) LRU : Least Recently Used

Remove data that was accessed the oldest.

2.) LFU : Least Frequently Used

3.) FIFO :- Remove the data that was written (into cache) the first/earliest

4.) LIFO

5.) MRU

6) MFU

⋮

There is NO absolute Best Algo.

Best algo depends upon your use-case

↳ Decide the best algo

Decide the best algo by what leads to the least # of cache miss.

Eg: Cache for IPL Match winners.

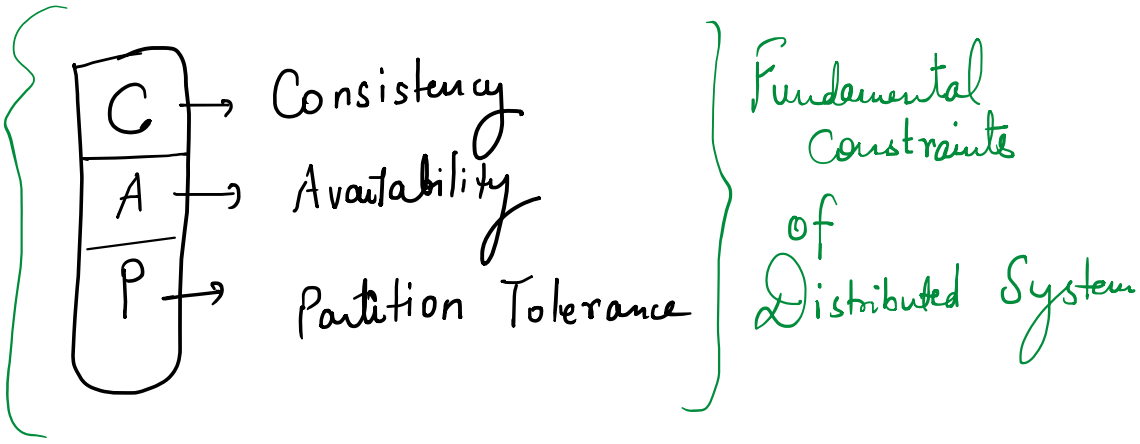
↳ FIFO.

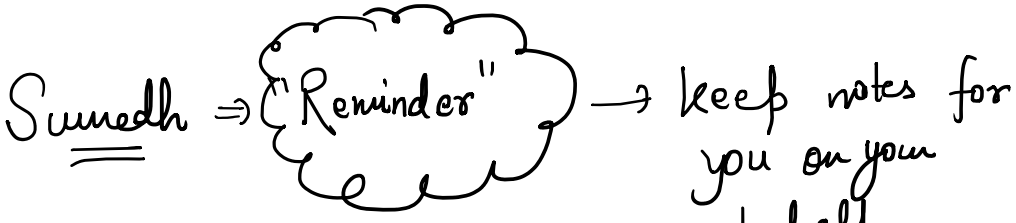
20%, 80%

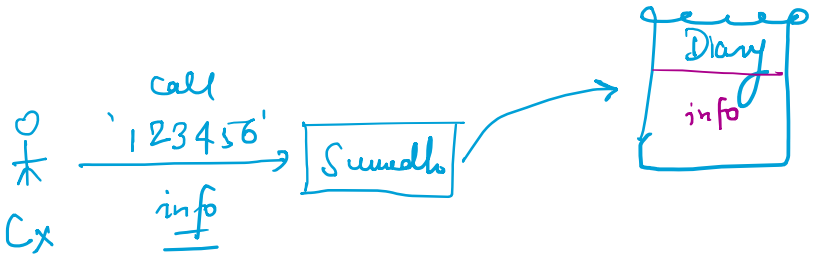
1:04pm

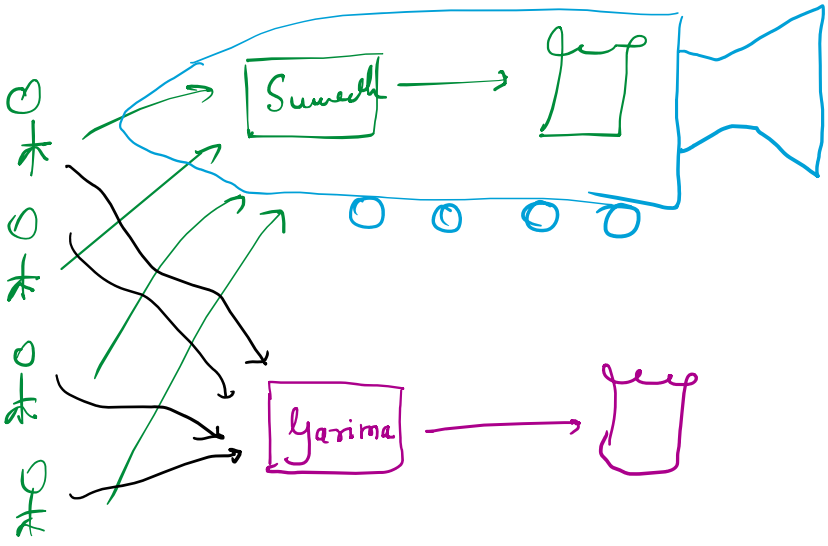
CAP Theorem & PACELC Theorem

↳ System Design 101

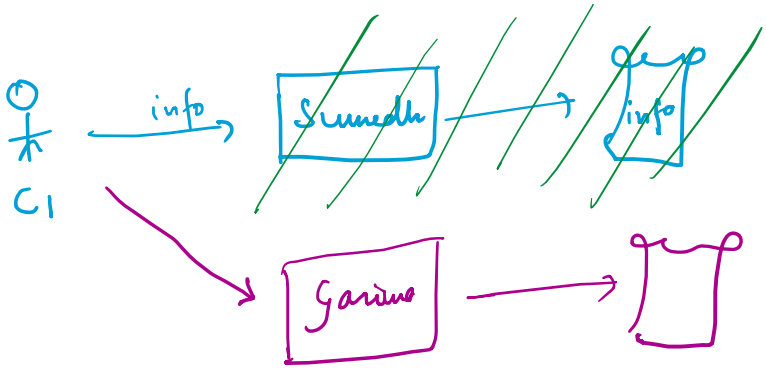

C
A
P
Consistency
Availability
Partition Tolerance
Fundamental Constraints of Distributed System


Sumedh
"Reminder"
keep notes for you on your behalf.


call
'123456'
info
Cx
Sumedh
Diary
info

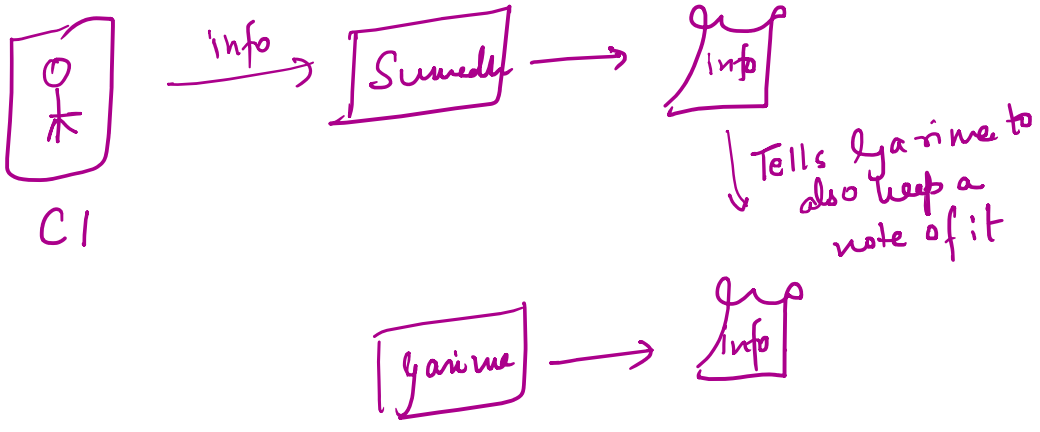

Sumedh
Yarima

Note :- Due to some obvious reasons, Sumedh & Gaurima can't just have a centralized diary. Hence, Gaurima has her own separate diary. :)



Gaurima wrongly feels ex C1 that he/she doesn't have any To-Do's.

Soln

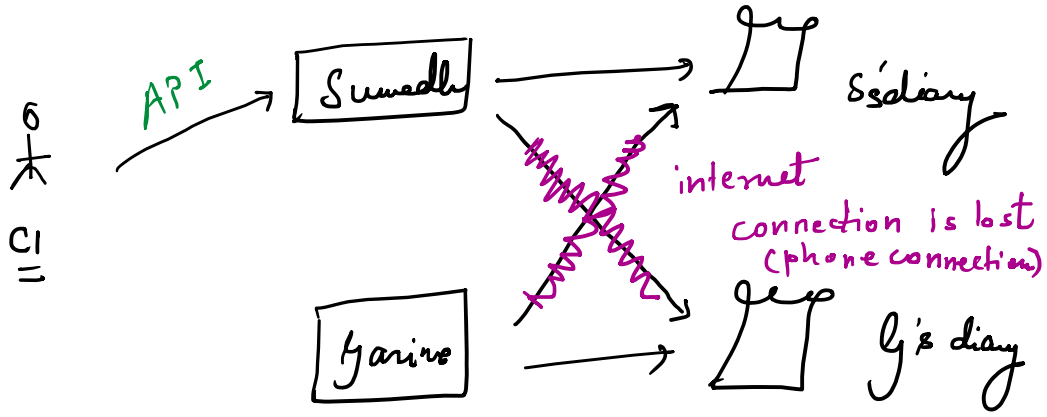


When ex tells any info to Sumedh, Sumedh puts it in his own diary AS-WELL-AS in Gaurima's diary that

AS-WELL-AS in journal

& then finally confirms to cx that notes has been stored.

Probl



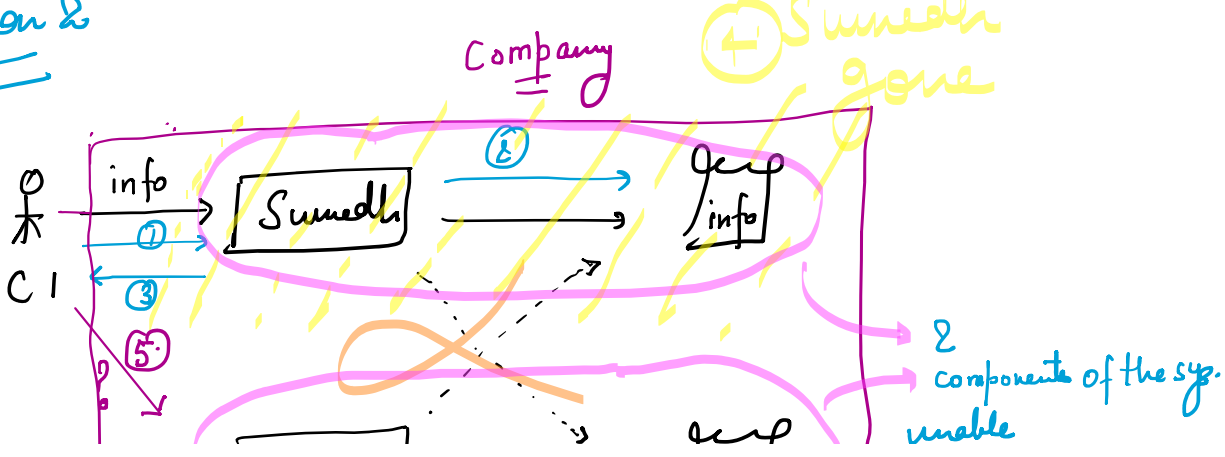
PS: Sumedh is unable to reach Gaurav (but API call [for writing info @ Sumedh's] needs to be replied back)

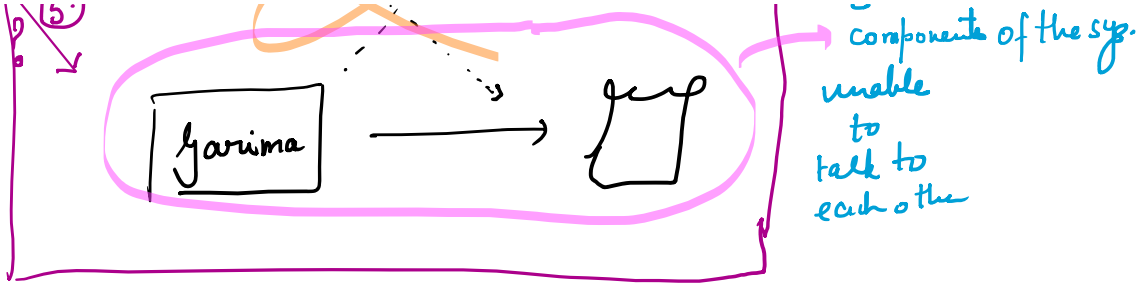
Soln (2 Approaches)

→ 1.) Tell cx I am unable to do this

→ 2.) Store only in your own diary & confirm to cx. And asyn write to other when inter-communication is back

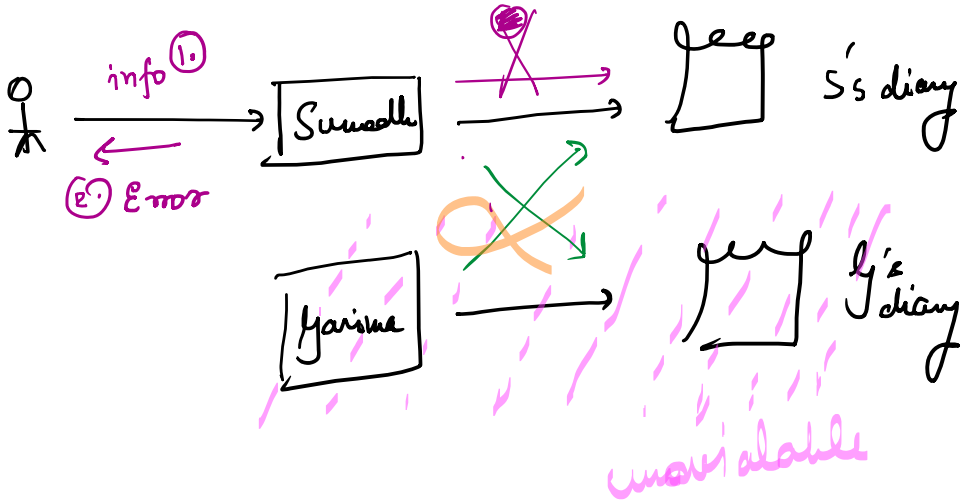
Option 2



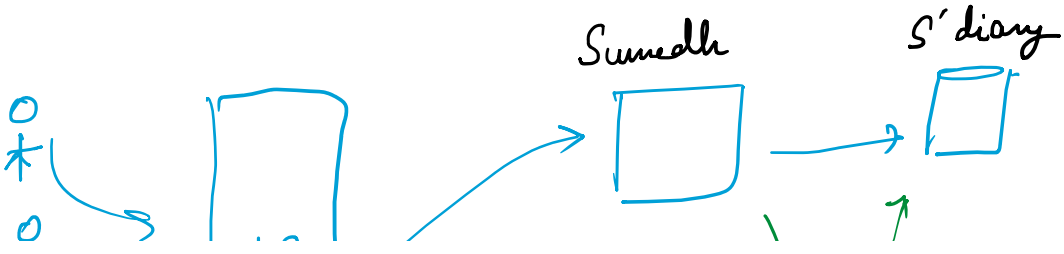


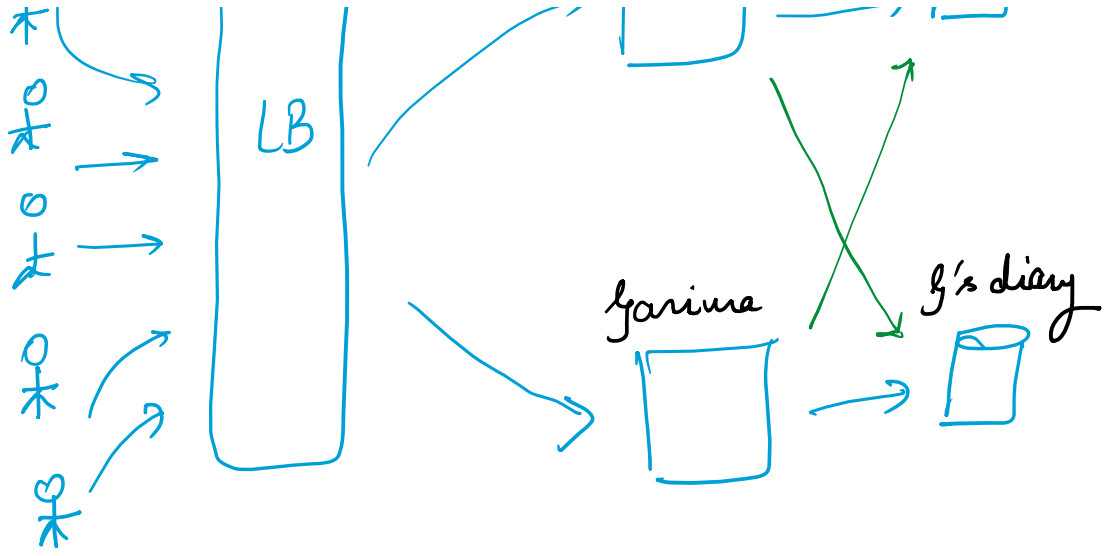
Consistency X Availability ✓

Option #1



Consistency ✓ Availability X





Appl^n Servers

Partition :- When s/w sys. gets divided such that there is no way to reach from one m/c to another

CA(P) → when a distributed sys. is internally partitioned & you want to exhibit partition tolerance (overall sys. is responding)

→ overall sys. can be available

OR

→ consistent

~~Ignore both~~

If it returns the ans, it is

Consistency :- If it returns the ans, it is guaranteed to be correct

BUT most of time it (sys) doesn't return anything at all

Availability :- will return, may or may not be correct

CAP :- Out of C, A, P, a sys. can has only 2 (never all 3 simultaneously)

impossible for a distributed sys.

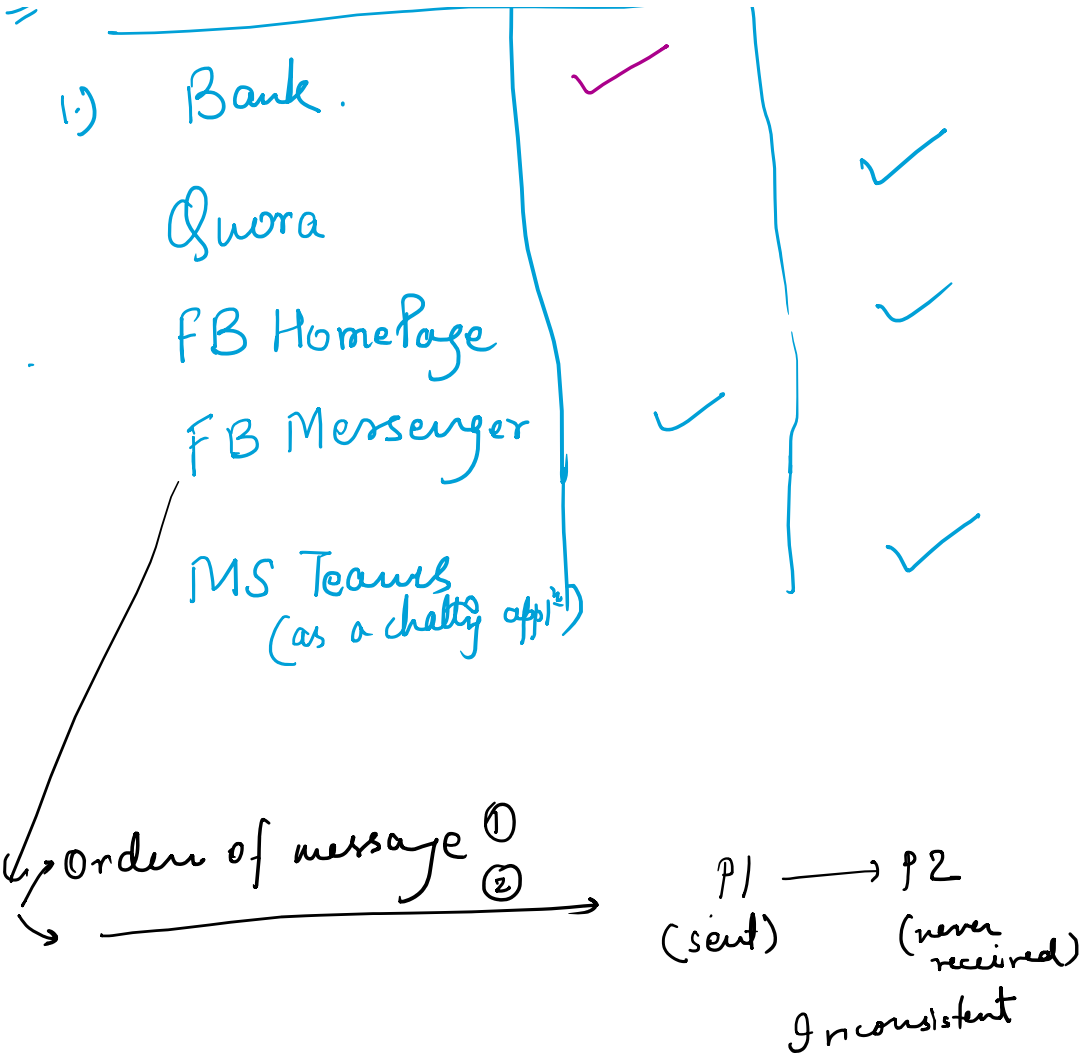
C, A → never partitioned (no ques. of partition tolerance)

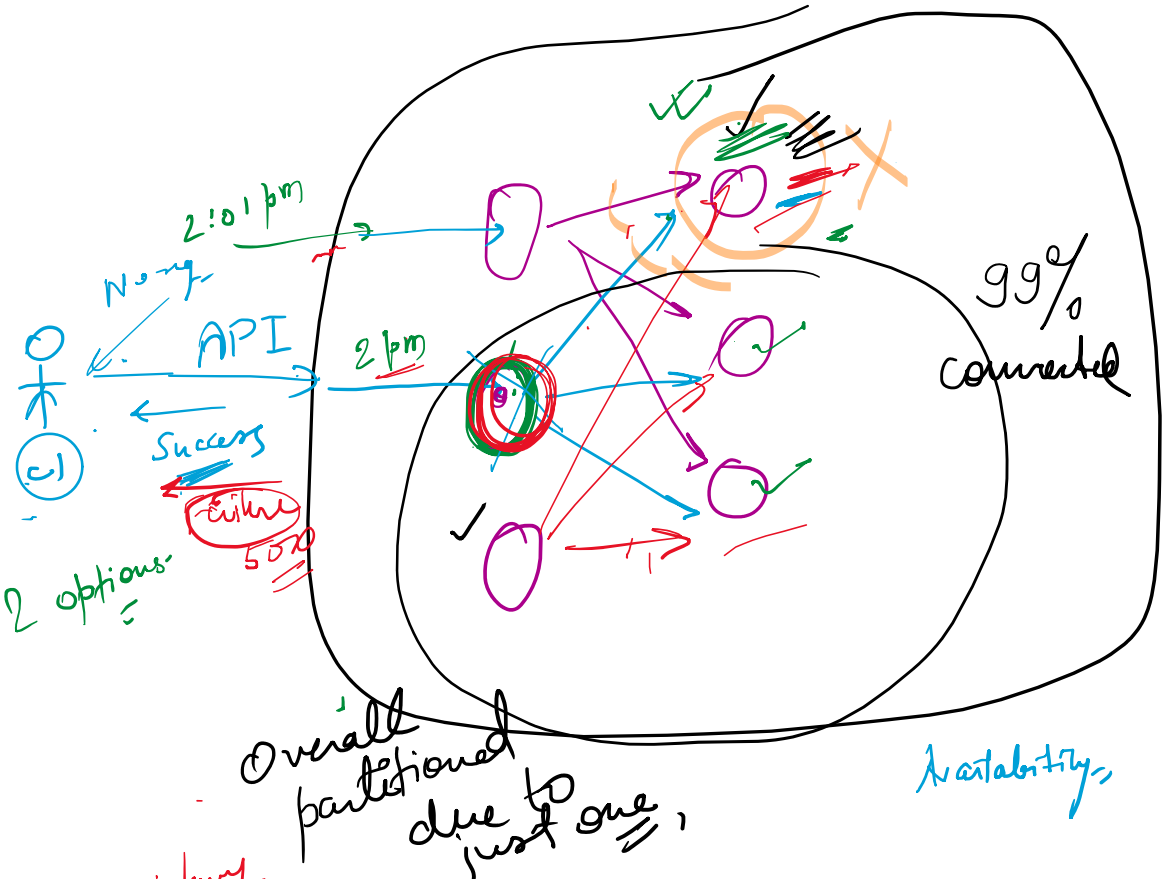
C, P → partition tolerance), consistency

A, P → " " , availability

* define consistency for each use case.

	C	A
Bank.	✓	


1.) Bank.
Quora
FB Homepage
FB Messenger
MS Teams
(as a chatty appl.)
order of message
1
2
P1
(sent)
P2
(never received)
Inconsistent


2:01 pm
Wrong
API
2 pm
Success
Failure
500
C1
2 options
99%
converted
Overall partitioned due to just one
Availability

Consistency